

Session 18 Quiz - AI in Everyday Life & Across Industries

Track B | Grades 9–12 | 20 Questions

Q1 [Type: MCQ | Difficulty: Easy | QID : 54522]

In healthcare, one of the most common uses of AI is _____ - analysing X-rays, CT scans, and MRIs to help doctors detect issues like lung nodules or tumours that might be missed by a tired human eye.

- A. Medical Imaging Analysis
- B. Algorithmic Trading
- C. De-aging
- D. Autocorrect

Correct Answer: A

Explanation: Medical imaging analysis is the headline healthcare AI application. CV models trained on millions of labelled scans can highlight subtle features - like a 3mm nodule - and flag them for radiologist review. Algorithmic trading is a finance application; de-aging is entertainment; autocorrect is NLP in messaging.

Q2 [Type: MCQ | Difficulty: Medium | QID : 54523]

In transportation, _____ uses AI to analyse real-time sensor data from engines, brakes, and other components in order to predict when a part is likely to fail before it actually does.

- A. Autocorrect
- B. Fraud Detection
- C. Predictive Maintenance
- D. Content Moderation

Correct Answer: C

Explanation: Predictive maintenance is the canonical transportation-AI application. The classic example: vibration data from a jet engine shifts in a way the ML model has seen three weeks before a specific bearing failed, so the part is replaced during the next overnight stop.

Q3 [Type: MCQ | Difficulty: Easy | QID : 54525]

In finance, AI fraud detection systems compare a new transaction against your normal spending patterns and produce a _____ - for example, "94%" - that the system uses to decide whether to flag or decline the purchase.

- A. Waveform
- B. Token count
- C. Pixel value
- D. Risk score

Correct Answer: D

Explanation: The fraud detection model outputs a numerical risk score that quantifies how unusual the transaction looks. A high score triggers a decline and an alert; a low score lets the transaction through quietly. Waveforms belong to Voice AI; tokens to NLP; pixels to Computer Vision.

Q4 [Type: MCQ | Difficulty: Medium | QID : 54527]

Across very different industries - healthcare, finance, transportation, entertainment, education - the underlying "AI Recipe" (Data → Algorithm → Training → Model → Prediction) is essentially the same.

- A. True
- B. False

Correct Answer: A (True)

Explanation: The data differs (X-rays vs. transactions vs. engine vibrations vs. film footage vs. quiz answers), but the recipe is the same: collect data, choose an algorithm, train on millions of examples, get a model, and use it to make predictions on new inputs. The surface tasks look very different, but the machinery is shared.

Q5 [Type: MCQ | Difficulty: Easy | QID : 54529]

When you unlock your phone with Face ID, the AI actually "knows" you as a person, in the same way a friend recognises you walking down the street.

- A. True
- B. False

Correct Answer: B (False)

Explanation: Face ID does not know you - it knows a mathematical pattern of your face is the most probable match. AI across every industry is pattern matching, not understanding. Convenience makes it feel personal, but Face ID is just a probabilistic match between your face's features and a stored template - no recognition in the human sense.

Q6 [Type: MCQ | Difficulty: Medium | QID : 54530]

Because AI tutors (such as Khanmigo, Socratic, or DreamBox) can analyse student work in real time and give personalised feedback, they can fully replace human teachers in every situation.

- A. True
- B. False

Correct Answer: B (False)

Explanation: AI tutors are excellent at scale and speed - spotting where a student went wrong, adapting practice, providing instant feedback. But teachers provide judgment, emotional support, encouragement, and the ability to read context AI misses (a student who is tired, anxious, has a speech difference, or is dealing with a difficult day). The core message is "AI + Humans" - AI handles scale and speed; humans provide wisdom, ethics, and empathy.

Q7 [Type: MCQ | Difficulty: Easy | QID : 54532]

Your phone's Portrait Mode blurs the background of a photo by estimating the depth of objects in the image. Which primary AI domain powers this?

- A. Natural Language Processing (NLP)
- B. Voice AI
- C. Recommendation systems
- D. Computer Vision

Correct Answer: D

Explanation: Estimating depth from a photograph is an image-analysis task, which falls under Computer Vision. It is the same family of techniques that powers face unlock, drone crop mapping, and medical imaging.

Q8 [Type: MCQ | Difficulty: Easy | QID : 54539]

In the entertainment industry, which of the following is a real example of how AI is used to create visual effects?

- A. Spam filtering for movie reviews
- B. Tokenising screenplays
- C. De-aging an actor
- D. Generating audiobook narration

Correct Answer: C

Explanation: ILM used machine learning trained on decades of younger footage to create a de-aged digital "mask" of Harrison Ford for flashback scenes. Tokenising screenplays is NLP infrastructure, not visual effects; audiobook narration is TTS, not de-aging; spam filtering is unrelated to visual effects.

Q9 [Type: MCQ | Difficulty: Medium | QID : 54540]

A smart-home thermostat reads the indoor temperature as 62°F at 6:45 AM, predicts the homeowner wants it 70°F by 7:00 AM, calculates that warming will take 12 minutes, and signals the furnace to start at 6:48 AM. Which Sense → Think → Act mapping is correct?

- A. Sense = reading the temperature; Think = signalling the furnace; Act = predicting the wake time
- B. Sense = predicting the wake time; Think = reading the temperature; Act = signalling the furnace
- C. Sense = reading the temperature and predicting; Think = calculating warm-up time; Act = signalling the furnace
- D. Sense = reading the temperature; Think = predicting and calculating warm-up time; Act = signalling the furnace

Correct Answer: D

Explanation: Sense is the raw data input (temperature reading). Think is the prediction and timing calculation. Act is the actual output to the world (the furnace signal). This is the same framework you have seen across NLP, CV, Voice AI, and Multimodal AI.

Q10 [Type: MCQ | Difficulty: Medium | QID : 54542]

Which option correctly matches an industry to an AI application?

- A.** Transportation → predictive maintenance of vehicle parts
- B.** Healthcare → algorithmic trading and high-frequency stock predictions
- C.** Finance → de-aging actors and creating visual effects for movies
- D.** Entertainment → drug discovery and protein structure prediction

Correct Answer: A

Explanation: Transportation is the industry where predictive maintenance is the canonical AI application - sensors stream data, an ML model predicts when a part will fail, and maintenance is scheduled before failure. The other options pair industries with applications that actually belong elsewhere.

Q11 [Type: MCQ | Difficulty: Medium | QID :54544]

The most successful AI systems are those where AI and humans work together. Which best captures that division of labour?

- A.** AI handles wisdom, ethics, and final decisions; humans handle the data crunching and pattern recognition
- B.** Humans should be completely removed from any process where AI is involved, as AI systems are now capable of handling judgment, ethics, and final decisions on their own
- C.** AI handles scale, speed, and pattern recognition; humans provide judgment, ethics, empathy, context, and final decisions
- D.** AI and humans do exactly the same things, just at different speeds and with different tools

Correct Answer: C

Explanation: AI is a pattern matcher; humans are the judges, ethicists, and supervisors. The radiologist confirming the AI's X-ray flag, the bank customer confirming the fraud alert, and the artist reviewing each de-aged frame all illustrate this division of labour.

Q12 [Type: MCQ | Difficulty: Medium | QID : 54545]

You are researching an AI industry topic and want to (1) get a broad overview quickly, and then (2) verify the specific facts against trusted sources you have uploaded. Which tool pairing makes sense?

- A. NotebookLM for broad exploration; Gemini for source-grounded verification
- B. Gemini for broad exploration; NotebookLM for source-grounded verification
- C. Gemini for both steps
- D. NotebookLM for both steps

Correct Answer: B

Explanation: Gemini is great for opening a topic up and seeing the landscape; NotebookLM grounds its answers in documents you upload, which lets you confirm specifics against trusted sources and reveals gaps when the source does not cover something - its "anti-hallucination" feature.

Q13 [Type: MCQ | Difficulty: Medium | QID : 54546]

A smart fridge notices you are low on milk and automatically adds it to your shopping list. Which AI domain is most likely doing the "sensing" of the milk level?

- A. Natural Language Processing
- B. Voice AI
- C. Algorithmic Trading
- D. Computer Vision

Correct Answer: D

Explanation: Cameras inside the fridge use Computer Vision to analyse visual scenes like the volume of a milk carton. NLP would handle written shopping lists, Voice AI would handle a spoken "add milk" command, and algorithmic trading is unrelated.

Q14 [Type: MCQ | Difficulty: Hard | QID :54548]

A logistics company uses AI to scan thousands of customer reviews and automatically sort them into "positive," "negative," and "neutral" categories. Which AI domain is doing the heavy lifting?

- A.** Computer Vision - it reads the visual layout of the review page
- B.** NLP - it analyses the meaning and tone of written text
- C.** Voice AI - it converts the reviews into speech before processing
- D.** Reinforcement Learning - it learns from customer rewards and punishments

Correct Answer: B

Explanation: Sentiment analysis is a core NLP task - the AI interprets the meaning and emotional tone of text. Computer Vision handles images, Voice AI handles audio, and Reinforcement Learning is a training method rather than a domain for analysing written reviews.

Q15 [Type: MCQ | Difficulty: Hard | QID :54549]

A music streaming app notices you play upbeat pop every weekday morning and lo-fi beats every weekend afternoon. One Saturday morning you feel anxious before a big exam and want energising music - but the app immediately queues the lo-fi weekend playlist. What does this best illustrate about AI recommendation systems?

- A.** The app is malfunctioning - it should detect your emotional state from your behaviour and adjust the playlist automatically to match your current mood and energy level
- B.** The app should ask you to rate your mood every morning so it can improve its predictions over time and learn to queue the right music for different emotional contexts
- C.** Recommendation systems identify patterns in past behaviour; they cannot detect your current emotional state, which is why human override remains essential
- D.** AI recommendation systems are generally unreliable and should not be used for personal suggestions about music, films, or any other type of content

Correct Answer: C

Explanation: Recommendation systems are pattern matchers trained on historical behaviour - they have no access to how you feel right now. The app is doing exactly what it was designed to do; the limitation is that past patterns cannot capture present context. This is why human control (skipping, overriding, adjusting) is a necessary part of any recommendation system's design.

Q16 [Type: MCQ | Difficulty: Hard | QID :]

A film studio uses AI to predict whether a movie will be profitable based on genre, budget, cast, and timing. An executive proposes using the AI to make final greenlight decisions on all new projects, removing the development team. What is the most significant problem?

- A.** AI predictions reflect past hits - films that break new ground or tell underrepresented stories don't match those patterns, so the AI would systematically reject them
- B.** The AI will become overconfident and start recommending increasingly expensive films, eventually pushing the studio toward financially risky productions it cannot sustain
- C.** The development team would have nothing to do, creating a workforce problem and damaging morale among the creative staff who built the studio's identity
- D.** Predicting film revenue is too complex for any AI model to handle accurately, since too many unpredictable cultural and market factors influence box-office success

Correct Answer: A

Explanation: Pattern-matching works best when the future resembles the past. But creative industries advance precisely when someone makes a film that does not look like previous hits - new genres, new voices, new formats. A model trained on historical success would have no basis for recognising innovation before it succeeds. The development team's human judgment is what evaluates what does not yet exist in the data.

Q17 [Type: MCQ Scenario | Difficulty: Medium | QID : 54551]

Scenario: You are on vacation in a foreign country and buy a small souvenir. Seconds later, a second transaction tries to buy a \$2,000 laptop online from your card. Within moments, your bank declines the laptop purchase and texts you an alert.

Question: Which industry and AI use is this?

- A.** Education; AI tutor
- B.** Finance; fraud detection
- C.** Entertainment; recommendation engine
- D.** Healthcare; medical imaging

Correct Answer: B

Explanation: This is the classic fraud-detection example. The AI does not "know" you are on vacation; it knows the statistical pattern of "foreign purchase followed immediately by high-value electronics" correlates strongly with stolen-card data, so it produces a high risk score and declines the suspicious transaction.

Q18 [Type: MCQ Scenario | Difficulty: Hard | QID : 54552]

Scenario: A drone flies over a cornfield and uses Computer Vision to create a "stress map" showing which sections of the field need more water or fertilizer. The farmer reviewing the map sees the AI has flagged a small patch as "stressed."

Question: Why must a human (the farmer) stay in the loop?

- A.** Because AI cannot make any reliable predictions about plants or crop conditions in agricultural settings
- B.** Because the drone cannot physically fly without human controls operating it from the ground at all times
- C.** Because AI is not legally allowed on farms and requires constant human supervision to comply with regulations
- D.** Because AI spots stress but cannot identify the cause - drought, disease, or pest - so a human must inspect

Correct Answer: D

Explanation: The CV system spots the symptom (stressed plants) but cannot diagnose the cause. A farmer's domain knowledge - inspecting the patch, considering recent weather, knowing what diseases are circulating - is what turns the AI signal into the right action.

Q19 [Type: FRQ - Short Answer | Difficulty: Hard | QID : 54553]

Scenario: A bank's AI detects a credit card transaction made at 2:00 AM, for \$500, from a device that has never been used before. Within seconds, the bank declines the transaction and sends a text alert to the cardholder.

Question: Walk through the Sense → Think → Act framework for this AI system. Then explain one reason why a human still needs to be in the loop, even though the AI handled the whole transaction in seconds.

(Respond in 3–4 sentences.)

Model Answer: In the Sense stage, the system collects the transaction data: amount (\$500), time (2:00 AM), merchant type, location, and the unfamiliar device. In the Think stage, an ML model compares this combination against the cardholder's normal spending patterns and computes a high risk score - say, 94%. In the Act stage, the transaction is declined and a text alert is sent to the cardholder. A human must stay in the loop because the AI does not actually know whether the purchase is legitimate - the cardholder might genuinely be travelling or buying a gift - so a person needs to confirm or override the decision and avoid blocking a real purchase, which is exactly the kind of judgment pattern matching cannot make on its own.

Q20 [Type: FRQ - Scenario Analysis | Difficulty: Hard | QID :54557]

Scenario: A city is considering deploying AI throughout multiple city services: automated translations for non-English speakers at government offices, AI scheduling for shelter intake, AI screening of housing voucher applications, and AI-powered transit navigation help for users with disabilities. The mayor argues this makes city services accessible to more people than ever. Disability rights advocates and civil liberties groups raise concerns about deploying AI across so many vital services simultaneously.

Question: Identify one significant benefit AND one significant risk of deploying AI broadly across multiple city services, and describe what the city should require to ensure these systems genuinely serve the residents they are meant to help.

(Respond in 3–4 sentences.)

Model Answer: A significant benefit is that AI can scale access to city services beyond what staff can manually provide - real-time translation for many languages, scheduling that works at any hour, navigation help on demand - particularly serving residents who currently struggle to access services because of language, time, or mobility barriers. A significant risk is that the same systems can systematically exclude or wrongly deny services to residents whose data patterns do not match what the AI was trained on, and because the systems are deployed across many essential services at once, a single resident could be wrongly screened out from housing, given inadequate translation, and unable to navigate transit help - compounding harm in ways hard to detect or contest. To deploy responsibly, the city should require independent audits of each system's performance across demographic groups before launch, accessible human appeal processes for every AI-driven decision, transparent disclosure of where AI is used and how, and an explicit commitment that AI cannot be the sole basis for denying anyone services - every denial requires human review with authority to override.